Networked Belief Formation and Portfolio Allocation

Endogenous Price Formation Under Topological Constraints

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Abstract

When an investor departs from an equal-weighted portfolio, they make a claim about which assets will outperform. This paper asks where such claims come from and how the structure of information networks affects whether they are correct.

We model investors as agents on a network who generate evidence about asset quality, share it with neighbours, and revise their beliefs accordingly. The shape of the network (who talks to whom) and its trust structure (how strongly agents discount evidence from those who disagree) jointly determine how quickly a group moves from ignorance to informed allocation.

Three results follow. First, network topology governs both the speed and reliability of convergence, independently of the number of connections. Second, when agents discount evidence from disagreeing neighbours, the effect on portfolio weights is algebraically equivalent to an increase in risk aversion. Third, when agents stop generating independent evidence and rely only on price and one another's behaviour, the market may still reach consensus, but that consensus need not reflect the underlying return structure.

An endogenous market model calibrated to FTSE 100 data (2008–2023) illustrates these dynamics across three regimes: the financial crisis, Brexit, and COVID-19.

Contents

1 Introduction and Definitions **2**
1.1 For readers from finance . . . 2
1.2 For readers from philosophy . . . 3

2 Belief Formation in Networks **3**
2.1 The setup . . . 3
2.2 Updating under full trust (Bayes) . . . 4
2.3 Updating under mistrust (Jeffrey) . . . 4
2.4 Three established results . . . 4

3 Network Topologies **4**
3.1 Complete network . . . 4
3.2 Cycle network . . . 5
3.3 Star network . . . 5
3.4 Scale-free (Barabási-Albert) network . . . 5

4 The Markowitz Objective **6**
4.1 The utility function . . . 6
4.2 Two-asset case . . . 6
4.3 Where does p_i come from? . . . 6

5 Mistrust as Effective Risk Aversion 7
5.1 Signal attenuation (α) . . . 7
5.2 Context matters . . . 7
5.3 Topology-dependent consequences . . . 7
5.4 Sensitivity of allocation to λ . . . 8

6 Credence Convergence and Portfolio Weight Dynamics 8

7 Polarisation and the Phantom Portfolio 9

8 Endogenous Price Formation 9
8.1 From beliefs to demand to price . . . 9
8.2 Exogenous and endogenous signals . . . 10
8.3 The simulation . . . 10

9 Regime Interpretation: FTSE 2008–2023 11
9.1 Financial crisis (2008–2009): hub fragility . . . 11
9.2 Brexit (2016): polarisation . . . 11
9.3 COVID-19 (2020–2021): endogenous feedback . . . 11
9.4 Summary . . . 11

10 Toward Endogenous Risk Aversion 11

11 Conclusion 12

1 Introduction and Definitions

This paper sits at the intersection of two frameworks: Markowitz portfolio theory (standard in finance) and epistemic network models (developed in philosophy of science). Readers from either side will find definitions from the other in this section.

1.1 For readers from finance

Agent. An individual decision-maker who holds beliefs, takes actions, and updates beliefs in response to evidence. In our models, each agent is an investor who allocates capital between two assets.

Credence. A number between 0 and 1 representing an agent's degree of belief in a proposition. A credence of 0.5 represents total uncertainty; 0.99 represents near-certainty. Credence replaces the binary "believe or do not believe" with a continuous measure of conviction. In portfolio terms, it is the agent's subjective probability that a given asset has positive excess return.

Bayesian updating (Bayes' rule). The standard rule for revising beliefs in light of new evidence. Given prior credence p and evidence e, the posterior credence is computed from the likelihood ratio of the evidence under competing hypotheses. The key property: evidence is taken at face value. If a neighbour reports experimental results, a Bayesian agent incorporates those results fully.

Jeffrey updating (Jeffrey's rule). A generalisation of Bayes' rule for situations where the agent is uncertain about the evidence itself. Rather than treating received evidence as certainly true, the agent assigns it a probability $P_f(e) < 1$. This allows for partial discounting of evidence. Bayes' rule is the special case where $P_f(e) = 1$.

Homophily. The tendency to trust people who are similar to oneself. In our models, similarity is measured by closeness of beliefs: agents discount evidence more heavily when it comes from neighbours whose credences differ substantially from their own. This is a specific assumption used to operationalise Jeffrey's rule; other implementations are possible.

Network topology. The shape of the communication graph connecting agents. Two networks can have the same number of connections (same density) but different shapes. A cycle (ring), a star (one hub connected to all others), a complete graph (everyone connected to everyone), and a scale-free graph (a few highly connected hubs, many peripheral nodes) all represent different topologies.

Exogenous and **endogenous.** An exogenous signal comes from outside the system: earnings data, economic indicators, experimental results. An endogenous signal is generated within the system: price movements, other agents' portfolio positions, trading volume. The distinction matters because exogenous signals carry information about the underlying return structure; endogenous signals may not.

Epistemic Relating to knowledge and belief. An epistemic process is one that produces, transmits, or revises beliefs. In this paper, "epistemic network" means a network through which agents form and update their beliefs about asset returns. The adjective distinguishes questions about what agents believe (epistemic) from questions about what they prefer (utility) or what they do (allocation).

1.2 For readers from philosophy

Portfolio weight. The fraction of an investor's capital allocated to a given asset. A weight of 0.5 in each of two assets is "equal-weighting" and represents a position of total agnosticism about which asset is better.

Risk aversion (λ). A parameter governing how much an investor penalises uncertainty. At high λ, the investor demands strong evidence before departing from equal-weighting. At low λ, even weak evidence triggers a large portfolio tilt. The parameter λ scales how aggressively belief translates into action.

Expected return. The mean payoff an investor anticipates from holding an asset. In our two-asset model, asset A has a known return of 0.5 (a coin flip), while asset B has a true return of $0.5 + \varepsilon$, where $\varepsilon > 0$ is the excess return. The agents do not initially know whether ε is positive. Discovering this is the epistemic task.

Markowitz optimisation. The standard framework for computing optimal portfolio weights given beliefs about expected returns, a covariance structure, and a risk aversion parameter. The key equation is given in Section 4.

2 Belief Formation in Networks

Before introducing portfolio theory, we describe the epistemic process that generates the beliefs portfolio theory takes as inputs. This section follows the simulation framework of Ball [8], building on Zollman [2] and O'Connor and Weatherall [3].

2.1 The setup

A group of N agents is arranged on a network $G = (V, E)$. Each agent holds a credence $p_i \in [0, 1]$ representing their belief that action B is superior to action A. In our models, B is in fact superior ($\varepsilon > 0$), but agents do not know this at the outset. Initial credences are drawn from a uniform distribution on $[0, 1]$.

Each time step, every agent whose credence exceeds 0.5 (i.e. who currently believes B is better) runs a binomial experiment: n independent trials, each succeeding with probability

$0.5 + \varepsilon$. The agent observes the outcome and shares it with their network neighbours. All agents then update their credences.

2.2 Updating under full trust (Bayes)

Under Bayes' rule, each agent treats the evidence received from neighbours as fully reliable. The posterior credence incorporates the likelihood ratio of the observed data under $\varepsilon > 0$ versus $\varepsilon = 0$. Convergence to the truth ($p_i \to 1$ for all i) is guaranteed in the long run, provided the network is connected and $\varepsilon > 0$.

2.3 Updating under mistrust (Jeffrey)

Under Jeffrey's rule, agents discount evidence from neighbours whose beliefs differ from their own. The final probability assigned to the received evidence is

$$P_f(E) = 1 - \min(1,\ d \cdot m)\,(1 - P_i(E)), \tag{1}$$

where $d = |p_i - p_j|$ is the absolute difference between the credences of the receiving agent i and the sending agent j, and $m \geq 0$ is the **mistrust multiplier**.

When $m = 0$: no discounting occurs and Jeffrey's rule reduces to Bayes' rule.

When $d \cdot m \geq 1$: the evidence is discarded entirely. The agent treats the neighbour's report as carrying no information.

This implements a form of homophily: agents trust those who agree with them and ignore those who do not. The parameter m controls how quickly trust decays with disagreement.

2.4 Three established results

The simulation framework yields three results relevant to what follows:

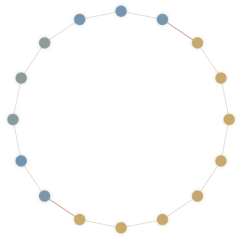
1. **Discounting delays convergence.** Even when all agents report truthfully, Jeffrey updating with $m > 0$ produces significantly slower convergence to the correct consensus than Bayesian updating (Mann-Whitney U, $p < 0.05$ across all tested parameter combinations) [8].
2. **Network shape affects convergence independently of density.** Networks of size 64 with density ≈ 0.03 produce mean convergence times of 68,818 steps (star), 109,127 (cycle), 119,976 (Barabási-Albert), and 120,596 (Watts-Strogatz), with all pairwise comparisons significant at $p < 0.005$ [8].
3. **High mistrust produces stable polarisation.** When $m > 1$, the network can fracture into subgroups that permanently disagree: one group converges to $p > 0.99$ while the other stabilises below 0.5, with credence distances too large for evidence to cross the gap.

3 Network Topologies

Four network types recur throughout each with a distinct information-flow profile.

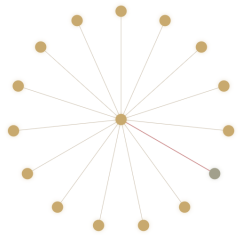
3.1 Complete network

Every agent is connected to every other. Information propagates instantly to the entire group. Convergence is fast, but the group is vulnerable to early misleading evidence: a few unlucky experimental draws can push the entire network toward a false consensus before corrective evidence accumulates. Zollman [2] identified this tradeoff; Rosenstock, Bruner, and O'Connor [5] showed it is most pronounced when the signal (ε) is weak and the network is small.

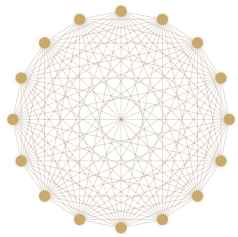


(a) Cycle network. Slow but robust propagation; disagreement remains locally contained.

(b) Barabási-Albert network. Hub concentration creates uneven information flow and visible epistemic stratification.



(c) Star network. Fastest convergence, but extreme vulnerability to hub error or isolation.



(d) Complete network. Maximal connectivity accelerates convergence amplifying correlated error.

Figure 1: Final network states for four topologies under identical simulation parameters. Node colour encodes terminal credence: gold indicates high credence in B, blue indicates commitment to A. Red edges mark high credence distance, signalling disagreement large enough to impede evidence flow under Jeffrey updating.

3.2 Cycle network

Each agent communicates with exactly two neighbours, forming a ring. Information must propagate around the entire ring to reach all agents. Convergence is slow. However, misleading evidence is diluted over many hops, making false consensus less likely. The cycle trades speed for reliability.

3.3 Star network

One central agent (the hub) connects to all others; peripheral agents connect only to the hub. Convergence is the fastest of all tested topologies because the hub aggregates evidence from every peripheral agent and broadcasts a consolidated signal back. The fragility is obvious: if the hub is misinformed, compromised, or removed, the entire network loses its only coordination mechanism. O'Connor and Weatherall [4] note that occupying the hub is a strategy available to disinformants.

3.4 Scale-free (Barabási-Albert) network

Generated by preferential attachment: each new node connects to existing nodes with probability proportional to their current number of connections. This produces a power-law degree distribution with a few highly connected hubs and many peripheral nodes. Information flow is uneven: agents near hubs update faster than those on the periphery. Ball [8] introduces a normalised measure of neighbourhood-size variance (moment_sd) and finds it correlates positively with convergence time.

4 The Markowitz Objective

With the epistemic framework established, we now introduce the portfolio theory that maps beliefs to allocations.

4.1 The utility function

The Markowitz mean-variance utility is

$$U = \mathbf{w}^\top \mathbf{E} - \lambda\, \mathbf{w}^\top \Sigma\, \mathbf{w}, \tag{2}$$

where $\mathbf{w}$ is the vector of portfolio weights, $\mathbf{E}$ the vector of expected returns, Σ the covariance matrix, and $\lambda > 0$ the risk aversion parameter.

The investor maximises U by choosing weights $\mathbf{w}$. The first term rewards higher expected return; the second penalises exposure to variance. The parameter λ controls the tradeoff: at large λ, even small variance is costly, and the investor stays close to a diversified baseline.

4.2 Two-asset case

Consider two assets. Asset A has a known, fixed probability of 0.5 of producing a positive outcome (a fair coin). Asset B has probability $0.5 + \varepsilon$, where $\varepsilon > 0$ is the true excess return. The agent does not know the value of ε; their credence p_i represents their subjective probability that $\varepsilon > 0$. The optimal weight in B is

$$w_B^* = \frac{1}{2} + \frac{p_i\, \varepsilon}{2\lambda\sigma^2}. \tag{3}$$

Three cases fix the interpretation:

- At $p_i = 0.5$ (complete uncertainty about whether B is better), the second term vanishes and $w_B^* = 0.5$. This is the equal-weighted portfolio: maximum ignorance expressed as allocation.
- As $p_i \to 1$ (near-certainty that B is superior), the weight tilts maximally toward B, with the magnitude constrained by λ and σ^2. Higher risk aversion or higher variance dampens the tilt.
- At $p_i < 0.5$ (the agent believes A is superior), the weight tilts toward A. The agent is allocating against the truth.

The key observation: every departure from equal-weighting is a knowledge claim. The size of the departure depends on the strength of belief (p_i), the magnitude of the perceived edge (ε), and the investor's willingness to act on uncertain information ($1/\lambda$).

4.3 Where does p_i come from?

In textbook Markowitz, the expected return vector $\mathbf{E}$ is an input. The investor "knows" the returns, or has a model that produces them. In practice, beliefs about expected returns are constructed through a social process: analysts generate estimates, share them with colleagues, revise in light of disagreement, and converge (or fail to converge) on a view. This is precisely the epistemic network process described in Section 2. The credence p_i is the output of that process, and the portfolio weight w_B^* is a deterministic function of it.

5 Mistrust as Effective Risk Aversion

This section establishes the formal equivalence between epistemic discounting and increased risk aversion.

5.1 Signal attenuation (α)

When agents discount evidence under Jeffrey updating, the credence that enters the portfolio optimiser is not the raw credence p_i but a reduced version. Define $\alpha \in (0, 1]$ as the **signal attenuation factor**: the fraction of available evidence that survives the discounting process. At $\alpha = 1$, all evidence is incorporated (Bayesian updating). At $\alpha < 1$, some evidence is lost to mistrust. The attenuated credence is αp_i. Substituting into equation (3):

$$w_B(t) = \frac{1}{2} + \frac{\varepsilon}{2\sigma^2} \cdot \frac{\alpha\, p_i(t)}{\lambda}. \tag{4}$$

This is algebraically identical to an unattenuated agent with a higher risk aversion:

$$\lambda_{\text{eff}} = \frac{\lambda}{\alpha}. \tag{5}$$

Since $\alpha \leq 1$, the effective risk aversion λ_{eff} is always at least as large as the intrinsic risk aversion λ. When α is small (heavy discounting, high mistrust), the effective risk aversion is much larger: the agent behaves as though they are far more cautious than their intrinsic preference would dictate. Their portfolio stays closer to equal-weighting for longer.

5.2 Context matters

The observed behaviour of an investor (their portfolio tilt, their trading frequency, their apparent caution) reflects not their intrinsic risk preference alone, but the combination of preference and epistemic environment. An investor embedded in a high-mistrust network looks risk-averse even if their intrinsic λ is low. Conversely, an investor in a high-trust, well-connected network translates belief into action more aggressively.

The two quantities (λ and α) cannot be disentangled from portfolio data alone without knowing the trust structure of the information network.

5.3 Topology-dependent consequences

The attenuation factor α depends on network position:

In a **star network**, peripheral agents receive evidence primarily from the hub, whose credence converges quickly to the group mean. Once convergence begins, d (credence distance) is small for hub-peripheral pairs, so discounting is minimal: $\alpha \approx 1$ and $\lambda_{\text{eff}} \approx \lambda$.

In a **cycle network** under Jeffrey updating, adjacent agents may hold different beliefs for extended periods. The credence distance d remains large, discounting is heavy, and α stays well below 1. The portfolio is pinned near equal-weight.

In a **Barabási-Albert network**, α varies across agents depending on their position relative to hubs. Agents near hubs (small d, low discounting) rebalance rapidly; peripheral agents (large d, heavy discounting) remain inert. This produces a heterogeneous distribution of effective risk aversion across the network.

5.4 Sensitivity of allocation to λ

The partial derivative of the optimal weight with respect to risk aversion is

$$\frac{\partial w_B}{\partial \lambda} = -\frac{p_i\,\varepsilon}{2\lambda^2\sigma^2}. \tag{6}$$

This derivative is convex in λ: at low λ (aggressive investors), small changes in risk aversion produce large changes in allocation. At high λ (cautious investors), the same perturbation produces a smaller effect. Since mistrust inflates λ_{eff} via α, the allocation error introduced by network structure is most consequential for agents who would otherwise be aggressive allocators.

6 Credence Convergence and Portfolio Weight Dynamics

The interactive simulation at `https://leelemacjames.github.io/ignorance-in-portfolios` implements the belief formation process described in Section 2 with the Markowitz mapping from Section 4 applied at each time step. The result is a joint visualisation: on the left axis, each agent's credence $p_i(t)$ evolves over time; on the right axis, the corresponding portfolio weight $w_B^*(t)$ follows from equation (3).

Figure 2 shows a representative run under a Barabási-Albert network with 16 agents. Several features are visible.

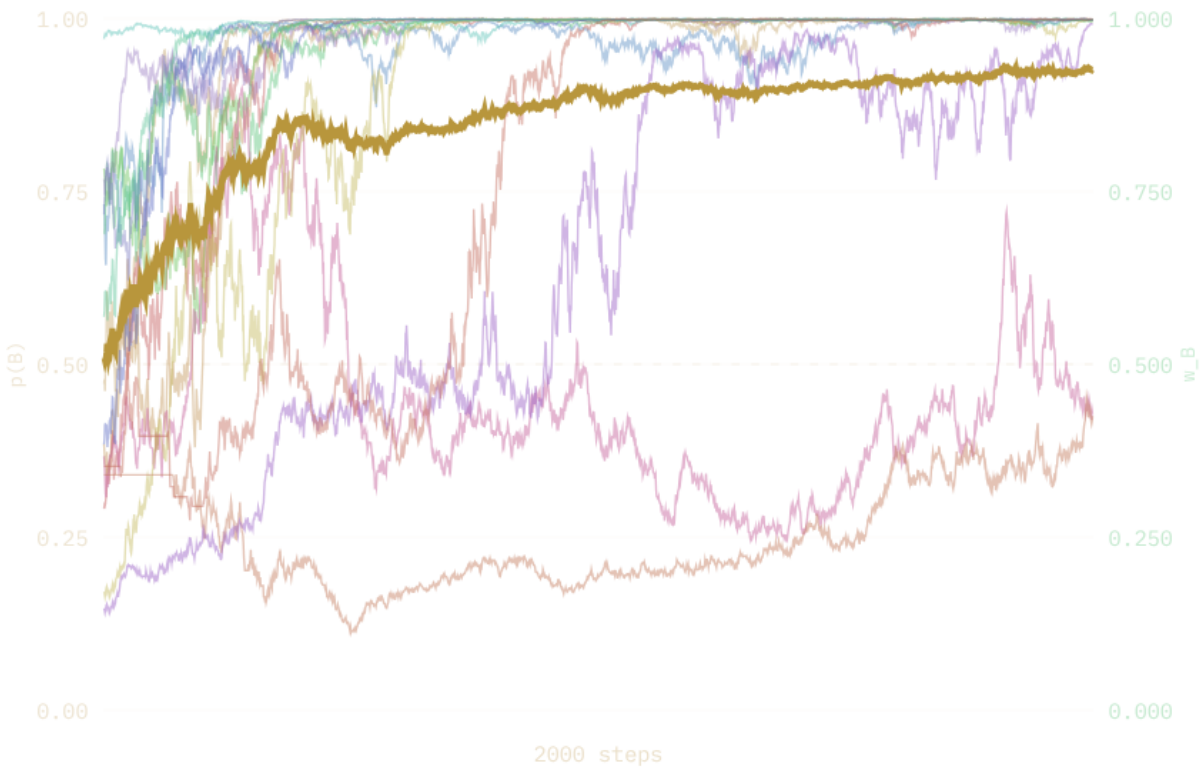


Figure 2: Credence and portfolio weight trajectories under a Barabási-Albert network with 16 agents. Each coloured path represents an individual agent's credence $p_i(t)$ that asset B is superior, while the bold gold line shows the population mean. The right axis shows the implied portfolio weight via the Markowitz mapping. Agents connected to highly central nodes update more rapidly; peripheral agents exhibit delayed adjustment.

First, agents do not converge at the same rate. Those with more network connections (closer to hubs) update faster, because they receive more evidence per step. Peripheral agents lag

behind. This is the topology effect from Section 3: the same density of connections produces different convergence dynamics depending on how those connections are distributed.

Second, the population mean (bold gold line) moves toward $p = 1$ (correct consensus) but does so unevenly, with periods of stagnation and rapid movement. Under Jeffrey updating, these plateaux correspond to episodes where credence distances between neighbours are large enough to trigger heavy discounting: evidence is being generated but not transmitted effectively across the network.

Third, the portfolio weight axis on the right makes the allocation consequence of slow convergence explicit. While the mean credence hovers near 0.6, the corresponding portfolio weight remains close to equal-weighting. The tilt toward asset B only becomes substantial once credence exceeds approximately 0.8. The delay between "the truth is available in the network" and "the portfolio reflects it" is the allocation drag imposed by epistemic structure.

The simulation allows users to vary topology, number of agents, ε, number of trials per step, update rule (Bayes or Jeffrey with selectable m), and risk aversion λ. A topology comparison mode runs all four network types with identical parameters and reports convergence time and reliability across 10 runs each, replicating the density-controlled comparisons from Ball [8].

7 Polarisation and the Phantom Portfolio

When the mistrust multiplier m exceeds 1, the network can fracture permanently. One subgroup converges to $p > 0.99$ (committed to B); the other stabilises below 0.5 (committed to A). The credence distances between the two groups satisfy $d \cdot m \geq 1$, so all cross-group evidence is discarded. Neither group updates in response to the other.

The aggregate portfolio weight across the full population,

$$\bar{w}_B = \frac{1}{N} \sum_i w_i, \tag{7}$$

is held by nobody. Every agent holds either a strong B tilt or a strong A tilt. The average is a statistical artefact: a **phantom portfolio** that reflects the capital balance between two irreconcilable camps rather than any shared estimate of ε.

In market terms, this corresponds to a persistent split between two non-communicating groups of investors. The market-clearing price is set by the balance of capital between them. It is not a transient disequilibrium that will be resolved by further evidence; it is a stable fixed point of the belief dynamics.

8 Endogenous Price Formation

Up to this point, the "price" of the asset has played no role: agents update beliefs from experimental evidence, and the portfolio weight is a private calculation. We now close the loop.

8.1 From beliefs to demand to price

When all agents translate their beliefs into portfolio weights, and those weights determine demand for the asset, the resulting price is an aggregate of the network's internal state:

$$P(t+1) = P(t) \cdot \left(1 + \eta \cdot \frac{1}{N} \sum_i (w_i(t) - 0.5)\right), \tag{8}$$

where $\eta > 0$ is the market impact parameter governing price sensitivity to excess demand. When agents collectively overweight the asset ($\bar{w} > 0.5$), price rises. When they underweight ($\bar{w} < 0.5$), price falls.

8.2 Exogenous and endogenous signals

Each agent receives signals that inform their belief. These signals divide into two kinds.

Exogenous signals originate outside the market. In the simulation, fundamental agents observe noisy measurements of the asset's true value (the FTSE close). These signals carry information about ε: they are evidence about the world.

Endogenous signals originate inside the market. Technical agents observe the market price and its recent history (moving average crossovers, momentum). These signals carry information about what other agents have done. They do not directly measure ε.

A network composed entirely of agents using endogenous signals has no mechanism for discovering the true value of ε. Such a network may converge to a consensus, but that consensus is a coordination equilibrium determined by internal dynamics. Convergence without exogenous evidence is convergence without knowledge.

8.3 The simulation

The accompanying interactive model[1] implements this framework using 3,784 days of FTSE 100 data (August 2008 to July 2023) as the exogenous fundamental value series. A Monte Carlo mode runs 20 simulations with identical parameters but different random seeds, producing a distribution of price paths whose spread measures the sensitivity of market outcomes to initial conditions and network randomness.

Figure 3: Endogenous market price versus fundamental value (FTSE 100) under a Barabási-Albert network with 32 agents and 50% technical traders. The green line represents FTSE closes (fundamental value); the gold line shows the simulated endogenous market price generated by networked belief formation. The shaded region highlights mispricing: deviation arising from network structure, mistrust, and the presence of self-referential agents.

[1] https://leelemacjames.github.io/endogenous-market.html

9 Regime Interpretation: FTSE 2008–2023

The three major episodes in the FTSE data correspond to three distinct configurations of the belief network.

9.1 Financial crisis (2008–2009): hub fragility

The crisis is characterised by the failure of centralised information sources: rating agencies, bank risk models, structured product pricing. In network terms, this corresponds to a star-like topology in which a small number of hubs dominate information flow. Under such a structure, convergence is fast (the hub anchors expectations) but fragile: when the hub's information is wrong, the error propagates system-wide before it can be corrected. The sharp, correlated drawdown in the FTSE is consistent with the collapse of a hub-dominated network that had converged rapidly to an incorrect consensus.

9.2 Brexit (2016): polarisation

The Brexit period introduces a different configuration. The issue is not a single corrupted hub but persistent disagreement between subgroups interpreting the same evidence differently. Under Jeffrey updating with $m > 1$, the network partitions into clusters that stop communicating. The aggregate portfolio weight $\bar{w}_B$ becomes a phantom portfolio: held by nobody, reflecting the capital balance between groups rather than a shared estimate. In the FTSE data, this appears as elevated volatility, abrupt reversals, and sensitivity to political signals.

9.3 COVID-19 (2020–2021): endogenous feedback

The pandemic represents a collapse in the availability of reliable exogenous evidence. Agents lack calibrated priors for the economic impact of a novel pathogen. Belief updating becomes predominantly endogenous: agents rely on price, policy announcements, and each other's behaviour. Network density increases (everyone watches the same signals), approximating a complete graph. Convergence is rapid but self-referential: price reflects the network's internal coordination rather than improved estimation of ε.

9.4 Summary

Regime	Topology	Mistrust	Outcome
2008–2009	Star-like (hub-dominated)	Low	Fast, fragile convergence; systemic error
2016	Fragmented / clustered	High	Polarisation; phantom portfolio
2020–2021	Dense / near-complete	Moderate	Rapid endogenous convergence

10 Toward Endogenous Risk Aversion

The preceding analysis suggests that λ should not be treated as a fixed parameter. If the epistemic environment modulates effective risk aversion, a natural specification ties λ to local disagreement:

$$\lambda_i(t) = \lambda_0 + \gamma \cdot \mathrm{Var}_{j \in N(i)}\big(p_j(t)\big), \tag{9}$$

where λ_0 is intrinsic risk aversion and the second term adds a penalty proportional to disagreement among the agent's neighbours. When neighbours agree, the agent invests at their natural aggressiveness. When neighbours disagree strongly, the agent becomes more cautious. This is a rational response to the observation that one's information sources are in conflict.

11 Conclusion

Three conclusions follow from embedding portfolio theory within an epistemic network framework.

First, network topology governs the speed and reliability of convergence from equal-weight ignorance to informed allocation. Star networks converge fastest but are vulnerable to hub corruption. Cycles are slow but robust. Scale-free networks produce stratified dynamics in which position within the network determines the speed of portfolio adjustment.

Second, epistemic mistrust is algebraically equivalent to increased risk aversion. The observed risk aversion of a market participant is a composite of their intrinsic preference and their epistemic environment: $\lambda_{\text{eff}} = \lambda/\alpha$. The two components cannot be separated from behaviour alone.

Third, when exogenous evidence weakens and agents rely on endogenous signals, the market can reach consensus without discovering the truth. Such consensus is coordination, not knowledge.

Allocation error is a joint function of risk aversion, mistrust, and network geometry:

$$\text{Allocation error} \sim f(\lambda, m, G). \tag{10}$$

Market efficiency, in this view, is a property of the epistemic architecture through which beliefs are formed.

Interactive epistemic network simulation:
https://leelemacjames.github.io/ignorance-in-portfolios
Endogenous market model:
https://leelemacjames.github.io/endogenous-market.html
Dataset:
https://www.kaggle.com/datasets/pavankrishnanarne/global-stock-market-2008-present
This paper:
https://leelemacjames.github.io/networked-belief-portfolio-allocation.pdf

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